

BREAK INTO AEROSPACE

NASA

Engineering Interview Preparation Guide

Targeted for the 2025–2026 Recruitment Cycle

NASA is the United States civil space agency, operating ten field centres plus JPL across human spaceflight, robotic exploration, Earth science, aeronautics research, and space technology development. Unlike every other organisation in this series, NASA hiring is governed by federal law and OPM rules — creating structural realities in the application and interview process that have no equivalent in private industry. This guide consolidates the preparation tasks, signal areas, and strategic context needed to compete effectively across all relevant engineering roles.

PART I · WHAT NASA ACTUALLY LOOKS FOR

By Seniority Level

- **Intern (OSTEM / Pathways)** — NASA has two distinct intern pipelines and they operate very differently. OSTEM internships are short-term and competitive on academic merit, research fit, and genuine mission interest — the interview is conversational, mentor-led, and focused on whether your technical background matches what a specific team needs right now. Intern interviews are mostly behavioural, relaxed in atmosphere, and focused on past experiences — with friendly engineers genuinely interested in the candidate rather than trying to screen them out. The Pathways Program is more formally governed: all Pathways job announcements go through USAJOBS, candidates complete USA Hire assessments measuring non-technical competencies, and the structured interview includes questions based on job-related competencies critical to NASA's Pathways positions. The Pathways interview is scored — every candidate gets the same questions evaluated against the same rubric. This is a legal requirement of federal hiring, not a preference. Attempting to charm your way around a structured federal interview question that has predetermined scoring criteria is a mistake. *Glassdoor, NASA*
- **Mid-Level (GS-11 through GS-13, ~2–8 years)** — At mid-level, NASA is hiring engineers who can contribute meaningfully to an active mission, programme, or research effort from day one. The filter is simultaneously technical depth in the relevant domain and demonstrated ability to operate within NASA's specific institutional framework — its review gates, its requirements management practices, its Technical Authority structure, and its knowledge-sharing culture. Interviews probe background knowledge of NASA's work, how experience aligns with the agency's mission, and past experiences with problem-solving under constraints. Mid-level candidates who have worked in commercial industry and never encountered government programme management, NPR (NASA Procedural Requirements) compliance, or the civil servant / contractor interface will have gaps that interviewers notice. *Glassdoor*
- **Senior (GS-14 and GS-15, 10+ years)** — At senior level, NASA is specifically looking for engineers who can serve as — or support — a Technical Authority. Three independent Technical Authorities were created after Columbia to provide independent input rather than relying only on programme managers to make decisions: the Chief Engineer, the Chief of Safety and Mission Assurance, and the Chief Health and Medical Officer. The current Chief of SMA is specifically focused on combating organisational silence and ensuring dissenting opinions can be heard. This is not a values statement — it is a governance struc-

ture with teeth. Senior engineers are expected to understand that their independent technical judgement has institutional protection and institutional expectation attached to it. They are also expected to bring multi-mission perspective, mentor the next generation of NASA engineers, contribute to lessons-learned databases, and interface with external partners, contractors, and sometimes Congress. *SpacePolicyOnline*

Hands-On Signal Areas

Domain	Core Competencies Expected
Prior NASA / Contractor Experience	Candidates who already understand the NASA environment — its review culture, its NPR framework, its contractor/civil servant boundary — require less onboarding. Former Pathways interns converting to full-time are especially preferred.
NASA SE Methodology & Tools	NASA systems engineering postings specifically name DOORS Next Generation, Jama, and Cameo/MagicDraw, and expect candidates to contribute to requirements analysis, architecture, trade studies, and verification — reflecting how NASA implements NPR 7123.1. <i>Fastweb</i>
Centre-Matched Domain Depth	NASA is highly specialised by centre: propulsion at MSFC, human spaceflight at JSC, Earth science at GSFC, aeronautics at LaRC, deep space at JPL. Candidates whose background maps directly to the target centre are significantly more competitive than generalist applicants.
Probabilistic Risk Assessment (PRA)	NPR 8705.5 governs PRA at NASA. Engineers who have used formal risk assessment methods — fault trees, event trees, PRA models — rather than qualitative risk registers alone are valued at mid and senior levels.
Human-Rating Awareness	For any programme involving crew, NPR 8705.2 governs human-rating requirements, covering all space systems hardware and software supporting human activity, with the goal of maximum reasonable assurance that failure will not result in crew fatality or permanent disability. Engineers from human-rated programmes understand a qualitatively different standard of verification evidence. <i>NASA</i>
Technical Writing & Documentation	NASA produces a prodigious volume of formal documentation — design documents, ICDs, test plans, anomaly reports, lessons-learned entries. Engineers who write clearly and precisely in formal technical formats are genuinely preferred over those who communicate only verbally or informally.
Publications / Patents / Conference Papers	Especially at research-oriented centres (GSFC, JPL, LaRC, ARC), evidence of contributing to the technical literature signals both communication ability and a disposition toward advancing the field, not just delivering a product.

How NASA's Interview Format Differs from Every Other Organisation in This Series

- **The structured interview is a legal requirement, not a stylistic choice.** Federal competitive service hiring requires that every candidate receive the same questions, scored against the same predetermined criteria, by the same panel. The structured interview lasts approximately 40 minutes and includes behavioural and situational questions based on job-related competencies. There is no room for interviewers to go off-script for a particularly interesting candidate. Attempting to extend answers beyond what the question asked, or pivot to topics not covered, works against you — every extra minute spent on the wrong competency is time not spent demonstrating the one being scored. *NASA*

- **USA Hire assessments precede the interview for many roles.** After applying, candidates may be invited to complete a USA Hire assessment evaluating non-technical skills such as attention to detail. These are online, timed, and scored — and the score determines whether you are referred to the hiring official at all. Candidates who ignore the assessment as a formality risk not reaching the interview. *NASA*
- **Veterans' preference is real and consequential.** Federal hiring law requires that veterans with qualifying service receive preference in selection. For a non-veteran competing against a veteran with equivalent scores, this is a structural disadvantage that cannot be overcome by a better interview performance. Understanding this before you invest significant prep time is important.
- **The USAJOBS resume format is not optional and is not a normal resume.** NASA encourages use of the USAJOBS resume format to ensure all important information is included, and all resumes must be two pages or fewer. Failing to follow the format, or submitting a designed PDF that the USAJOBS system cannot parse, can result in automatic disqualification — a risk that does not exist in private-sector hiring. *NASA*
- **JPL is a special case.** JPL is managed by Caltech under contract to NASA and is an FFRDC — engineers there are Caltech employees, not federal civil servants. JPL onsite interviews typically consist of several one-on-one interviews with group heads and a technical presentation to the department — a full-day process from 8am to 4:30pm. The process is more similar to a research institution or university than to a federal agency interview. The civil-service-specific advice in this guide (USA Hire, structured interview, veterans' preference) does not apply to JPL — but the technical depth, research orientation, and mission-culture advice does. *Glassdoor*
- **Technical depth in interviews is real but context-dependent.** Phone interviews have covered both engineering behavioural questions and systems technical probing — for example, walking through components of a rocket engine and different types of systems. The technical depth varies significantly by centre and team. Research-heavy teams at GSFC or JPL probe more deeply; operational teams at KSC or MSFC may focus more on process and experience. *Glassdoor*

Honest caveat on format variation: NASA has ten field centres plus JPL, each with distinct missions, cultures, and hiring practices. A structures engineer at MSFC working on SLS experiences a completely different daily environment from an Earth scientist at GSFC or a flight operations engineer at JSC. Interview prep that works well for one centre may not map precisely to another. Filter Glassdoor and LinkedIn reviews by centre and specific directorate wherever possible.

PART II · 20 SPECIFIC PREPARATION TASKS

Intern Level (OSTEM and Pathways)

1. Build a USAJOBS-Compliant Resume Before Doing Anything Else — and Verify It Parses Correctly 3–4 h

Skill: Federal application process compliance

Objective: NASA job opportunities are posted on USAJOBS and the agency encourages using the USAJOBS resume format to ensure all important information is included. A resume that does not include the specific information USAJOBS needs (month and year of employment, hours per week, supervisor names for relevant positions, GS grade equivalent for prior federal work) can result in a candidate being scored as unqualified regardless of their actual experience. This is a filter that eliminates strong candidates before any human reviews their application.

Done Signal: You have a complete USAJOBS-format resume, verified it uploads and parses correctly

in the USAJOBS system, and confirmed that each position entry includes all required fields. A colleague who does not know your background has read it and can name your three strongest technical qualifications from it alone.

Source: NASA

3–4 h

2. Complete the USA Hire Practice Assessments and Understand How They Are Scored

Skill: Non-technical competency assessment performance

Objective: USA Hire assessments measure non-technical competencies critical to success in NASA's Pathways positions and results are used to place candidates into qualified, highly qualified, or best qualified categories — determining whether they are referred to the hiring official at all. Candidates who treat these as formalities and rush through them can score themselves out of consideration before a human ever sees their name.

Done Signal: You have completed at least two full-length USA Hire practice assessments (available through OPM's website), reviewed every question where your answer did not match the preferred answer, and can articulate the competency each question type is measuring.

Source: NASA

3–4 h

3. Prepare STAR Answers Mapped to the Specific Competencies Listed in Your Target Job Announcement

Skill: Competency-specific behavioural interview preparation for a scored structured format

Objective: NASA's structured interview scores answers against pre-defined competencies drawn from the vacancy announcement. The structured interview includes questions based on job-related competencies critical to the specific position. Generic STAR preparation misses this entirely. The competency listed in the vacancy announcement is not decorative — it is the scoring rubric. An answer that does not directly address the named competency earns a lower score regardless of how impressive the story is.

Done Signal: You have read the full vacancy announcement for your target position, identified every competency named in the assessment or interview section, and written one STAR story per competency, tagged to it explicitly. Each story has been spoken aloud to confirm it directly addresses the competency in the first 30 seconds.

Source: NASA

4–5 h

4. Prepare a Specific, Technically Grounded Answer to “Why NASA?” — Connected to a Centre, a Programme, and a Scientific or Engineering Problem

Skill: Mission and centre-specific conviction articulation

Objective: NASA interviewers distinguish between candidates who want to work somewhere meaningful and candidates who want to work at NASA specifically — on a specific mission, with a specific team, in a specific domain. Interviews probe knowledge of NASA's work and how experience aligns with the agency's mission. A “Why NASA?” answer that could apply to any space employer signals that the candidate has not engaged with what NASA specifically does and why it requires a civil servant workforce rather than contractors.

Done Signal: You have a 90-second answer that names: the centre you are applying to and what it is specifically known for, one current or recent mission or programme at that centre, one specific engineering or scientific challenge that mission involves, and how your background connects to that challenge. You cannot use the words “dream,” “inspire,” or “explore” as load-bearing terms.

2–3 h

Source: Glassdoor

2–3 h

5. Understand the Distinction Between NASA Civil Servants and Contractors — and What It Means for Your Role 2 h

Skill: Institutional context literacy

Objective: NASA centres are populated by both civil servants and contractor employees who work side by side on the same programmes. Civil servants make technical authority decisions; contractors execute work under their oversight. Interns who do not understand this boundary — who assume everyone in the building is a NASA employee, or who do not understand why their future civil servant role has different responsibilities and constraints than a contractor's — will be visibly unfamiliar with the environment they are joining.

Done Signal: You can explain in plain language the difference between a NASA civil servant and a support contractor, why the distinction matters for technical decision-making authority, and what the civil servant role specifically owns that a contractor cannot. You can do this without reading it off a page.

Source: APPEL Knowledge Services

6. Refresh Core Technical Fundamentals Relevant to Your Target Centre's Mission Domain 5–6 h

Skill: Domain fundamentals applied to NASA's specific technical context

Objective: While NASA intern interviews lean heavily behavioural, technical probing does occur — covering components of rocket engines, types of space systems, and domain-specific engineering concepts. The questions are conversational rather than whiteboard-based, but the expectation is that an engineering candidate has solid fundamentals in their declared domain (orbital mechanics, structural analysis, fluid systems, embedded software, GNC, thermal engineering) appropriate to their level of study.

Done Signal: For five technical topics directly relevant to your target centre and role (e.g., if targeting MSFC propulsion: combustion efficiency, specific impulse, turbopump design, propellant management, nozzle expansion), you can explain the core concept, give a NASA-relevant example, and answer one follow-up question without hesitation.

Spread over multiple days. Source: Glassdoor

Mid-Level (GS-11 through GS-13)

All intern tasks still apply. The STAR stories must be from professional contexts. The USAJOBS resume (task 1) must include full job history with all required federal fields completed. The "Why NASA?" answer (task 4) must connect to the specific directorate and branch, not just the centre.

7. Study NPR 7123.1 (NASA Systems Engineering Processes and Requirements) at an Operational Level 5–6 h

Skill: NASA's governing systems engineering framework

Objective: NPR 7123.1 establishes systems engineering processes and responsibilities across NASA programmes and projects. This is the governing document for how NASA engineers decompose requirements, develop architectures, manage interfaces, and verify designs. Mid-level candidates who come from industry with strong systems engineering backgrounds but have never read the NASA SE framework will find that the terminology, review gates, and process structure differ meaningfully

from commercial practice. NPR 7123.1 is publicly available.

5–6 h

Done Signal: You have read the executive sections of NPR 7123.1, can name the seventeen systems engineering processes it defines, explain the role of the System Engineering Management Plan (SEMP), and describe how NASA's lifecycle reviews (SRR, SDR, PDR, CDR, SAR, ORR) map to the V-model structure. You can do this without notes.

Source: NASA

8. Understand NASA's Technical Authority Structure and What It Means for an Engineer's Daily Work

3 h

Skill: Independent Technical Authority operational literacy

Objective: The three Technical Authorities — the Chief Engineer, the Chief of Safety and Mission Assurance, and the Chief Health and Medical Officer — were created after Columbia to provide independent input, with the specific goal of combating organisational silence and ensuring dissenting opinions can be heard. At mid-level, understanding the TA structure means knowing that you have both the right and the institutional obligation to raise a technical concern through TA channels even if programme management prefers otherwise. Mid-level engineers who do not know this structure exists are missing a fundamental piece of how NASA governance works.

Done Signal: You can explain the three Technical Authorities, why they were created, what authority they have to stop or modify a programme, and how an individual mid-level engineer's technical concern reaches them. You can give a real or hypothetical example of a situation where using the TA pathway would be appropriate versus where a normal programme management escalation would suffice.

Source: SpacePolicyOnline

9. Prepare a Story About Working Within a Formal Review Process — Where the Review Caught or Changed Something Real

2 h

Skill: NASA lifecycle review culture participation

Objective: NASA's programme review cycle — SRR through ORR — is not a box-ticking exercise. It is the primary mechanism by which technical risk is surfaced and resolved before flight. Mid-level engineers are expected to prepare for, present at, and defend designs in formal reviews. Candidates who have only been through informal peer reviews or design critiques have not experienced the level of preparation, documentation, and challenge that a NASA programme review involves.

Done Signal: You have a story involving a formal review (PDR, CDR, design review, safety review, or equivalent) where the review process itself identified a gap or forced a design change. The story names the specific finding, how it was dispositioned, and what was different about the design or test plan as a result. "The review went smoothly" is a weak answer.

10. Build a Working Knowledge of NASA's Risk Management Framework — Specifically NPR 8000.4 and PRA Concepts

3–4 h

Skill: Formal risk management using NASA's classified risk approach

Objective: NPR 8000.4, Agency Risk Management Procedural Requirements, is a governing document referenced across NASA's SE, software, and safety standards. NASA uses a specific risk classification system (likelihood × consequence on a 5×5 matrix) and distinguishes between programme risk (cost, schedule, technical performance) and safety risk. Mid-level engineers on flight programmes are expected to identify, document, track, and mitigate risks using this framework, not

just describe risks in narrative form.

3–4 h

Done Signal: You can explain NASA's 5×5 risk matrix, the distinction between likelihood and consequence ratings, how a risk is tracked through its lifecycle (open, mitigated, closed, accepted), and the difference between a risk and an issue. For one real project you have worked on, you can reconstruct a risk register entry with all required fields.

Source: NASA

11. Prepare a Story Demonstrating How You Navigated Ambiguity in Requirements — with a Documented Outcome 2 h

Skill: Requirements clarification and formal change management

Objective: NASA requirements — especially at programme level — are often ambiguous, contradictory between levels, or evolving as mission design matures. Mid-level engineers are expected to recognise ambiguity, seek clarification through the correct channel (not just interpret and proceed), and document the resolution. Candidates who “made a reasonable engineering judgement and moved on” without documenting the ambiguity resolution are describing a practice that creates significant risk in a NASA context.

Done Signal: You have a story where you identified an ambiguous or conflicting requirement, formally documented the ambiguity, sought clarification through an appropriate channel (requirements review, interface control board, chief engineer review), and the outcome is recorded in a traceable document — not just an email or a verbal agreement.

12. Study the NASA Systems Engineering Handbook (SP-2016-6105) — Specifically the Sections on Trade Studies, Interface Management, and Verification 5–6 h

Skill: NASA SE Handbook as operational reference

Objective: The NASA SE Handbook is the practitioner's companion to NPR 7123.1. It provides the *how*, where the NPR provides the *what*. Mid-level candidates who can reference specific SE Handbook guidance in an interview — citing the trade study methodology, the interface management approach, or the verification matrix structure — demonstrate they have done real systems engineering at the NASA-standard level, not just described systems engineering from a textbook.

Done Signal: You have read the sections on trade studies, interface management, and technical performance measures (TPMs) from the NASA SE Handbook, and for each section can describe what the document recommends and give a personal example of either using that approach or a situation where using it would have improved an outcome.

13. Prepare a Story Demonstrating Knowledge Management — Specifically Contributing to Lessons Learned or Technical Knowledge Transfer 2–3 h

Skill: Institutional knowledge preservation as an engineering discipline

Objective: The lessons of Columbia remind NASA that even those with the greatest technical acumen need a supporting culture to apply and share their knowledge effectively. NASA's people must combine the facts they know, the lessons they learn as a team, and the common values they practise together. NASA's APPEL Knowledge Services exists specifically to prevent tacit knowledge from being lost when engineers retire or programmes end. Mid-level engineers who have never written a lessons-learned entry, contributed to a post-mission review, or documented a technical decision for future reference are missing a practice that is institutionally central at NASA.

Done Signal: You have written at least one formal lessons-learned entry (for any project, in any

format) covering: what happened, what the root cause was, what was done differently as a result, and what future programmes should know. You can explain why lessons-learned documentation is different from a project post-mortem.

Source: NASA

2–3 h

Senior Level (GS-14 and GS-15)

All mid-level tasks still apply. NPR 7123.1 (task 7) must be operational knowledge — you should have implemented it, not just studied it. The TA structure (task 8) must include personal experience exercising or supporting independent technical authority. Risk management (task 10) must include PRA participation, not just qualitative risk registers.

14. Prepare to Describe a Time You Exercised or Supported Independent Technical Authority Against Programme Schedule Pressure

2–3 h

Skill: Technical Authority in practice — the senior differentiator at NASA

Objective: This is the question that matters most at senior level and that has no direct equivalent at any private employer. The Columbia accident report concluded that rigid organisational and hierarchical policies were preventing the free and effective communication of safety concerns, and that rank and stature were trumping expertise. NASA created the TA structure precisely to prevent this. Senior engineers are expected to have personally exercised or supported independent technical authority — raised a concern that slowed or stopped a programme, documented a dissenting position, or used the TA escalation pathway. Candidates who have not done this, or who frame every safety concern as having been resolved by consensus, are demonstrating they have not operated at the level NASA needs at GS-14+.

Done Signal: You have a story where you or a TA you supported stopped or significantly modified a programme decision on technical or safety grounds, the programme manager's preference was different from the TA finding, the concern was documented formally, and the outcome is traceable. The story names the specific technical concern and the specific document that captured the resolution.

Source: AICbE

15. Prepare to Walk Through a Full Programme Lifecycle — from Formulation Through Closeout — Including Every Formal Review Gate

4–6 h

Skill: End-to-end NASA programme ownership narrative

Objective: Senior NASA engineers are expected to have owned a significant technical thread across a full mission lifecycle — not just the development phase. The specific NASA review structure (Concept Review → SRR → SDR → PDR → CDR → SAR → ORR → Mission Operations Review → Closeout) maps to specific technical deliverables and maturity standards. Candidates who can walk through this structure from personal experience — naming what they were responsible for at each gate, what the hardest closure was, and what lessons they brought to the next programme — are demonstrating senior-level depth that candidates who know the gates only abstractly cannot match.

Done Signal: For one programme you were personally involved in, you can answer questions at every review gate without notes: what the open items were, who held the action, what the resolution was, and whether the programme passed or was conditionally approved. You can also name what you would do differently at two or three of the gates.

16. Prepare Examples of Influencing Contractor Technical Performance as a Civil Servant — Including a Case Where You Had to Hold a Contractor to a Requirement They Disputed 2 h

Skill: Civil servant oversight of contractor technical work

Objective: Senior NASA civil servants are not doing the engineering themselves in most cases — they are setting technical direction, reviewing contractor deliverables, and holding contractors to requirements. The ability to evaluate a contractor's technical submission, identify a gap against the requirement, document the finding, and hold the line through a formal response process is a core senior civil servant function. Candidates who have only worked as individual contributors, or who have never been on the government side of a contractor interface, are missing this competency.

Done Signal: You have a story where you reviewed a contractor's technical deliverable (design document, analysis report, test plan, or equivalent), identified a non-compliance or gap against a requirement, documented it in a formal review comment or finding, and tracked its resolution through closure. The story names the requirement, the gap, and the formal document in which the finding was recorded.

17. Prepare to Discuss a Probabilistic Risk Assessment (PRA) You Have Contributed To — Including How It Changed a Design or Operations Decision 4–5 h

Skill: Formal PRA contribution and decision influence

Objective: NPR 7123.1 and NPR 8705.5 govern technical probabilistic risk assessment procedures for safety and mission success across NASA programmes. PRA is used for crew safety analysis, mission reliability estimation, and risk-informed design trade-offs. Senior engineers on human-rated programmes are expected to understand PRA structure — fault trees, event trees, initiating events, basic events, uncertainty quantification — and to have used PRA results to inform at least one real engineering or operations decision. Describing PRA as something the safety office does is not the senior answer.

Done Signal: You can describe a PRA you contributed to, name the specific fault tree or event tree you worked on, explain how the top event and basic events were defined, and describe at least one design decision or operations constraint that resulted directly from the PRA analysis. You can explain what an importance measure (Fussell-Vesely, risk achievement worth) tells you and why it matters for prioritising mitigations.

Source: NASA

18. Prepare Examples of Building Technical Capability in a Team — Specifically Preparing a Junior Engineer to Make Independent Technical Decisions on the Programme 2 h

Skill: Technical leadership and knowledge transfer for a mission-critical context

Objective: NASA's workforce age profile at many centres skews toward retirement eligibility, and the institutional knowledge at risk is irreplaceable for ongoing programmes. The Columbia lessons remind NASA that even technically excellent individuals need a supporting culture and knowledge-sharing practice to apply their expertise effectively. Senior engineers who have actively transferred programme-specific knowledge — not just domain knowledge — to junior colleagues are performing a function that is central to NASA's long-term mission success. The specific question probes whether the junior engineer can now operate independently in a NASA context, not just in a generic engineering context.

Done Signal: You have a story where a junior engineer could not independently perform a NASA-specific function (writing a compliant requirements document, preparing for a programme review, applying a risk management framework, using a TA escalation pathway) and after your mentoring they could. The story names the specific capability, the method you used to transfer it, and the programme artefact that demonstrated the transfer was successful.

Source: NASA

2 h

All Levels

19. Prepare a Specific Answer to “Describe a Time You Raised a Concern That Others Were Normalising as Acceptable”

2 h

Skill: Anti-normalisation-of-deviance behaviour — the deepest cultural probe at NASA

Objective: Both Challenger and Columbia investigations found that conditions inconsistent with NASA’s technical basis for mission safety were being normalised — treated as acceptable because they had not caused a problem yet. Rank and stature were trumping expertise, and rigid hierarchies were preventing the communication of safety concerns. NASA asks this question at every level because normalisation of deviance is the specific failure mode its entire safety culture is designed to prevent. The answer must involve a specific technical anomaly or deviation, a specific person or group who had accepted it as routine, and a specific action you took to flag it — even if the outcome was not what you wanted.

Done Signal: You have a story where you identified something being treated as normal that was not normal — an anomaly, a process deviation, an assumption that was no longer valid — raised it explicitly against resistance or inertia, and can name both the technical concern and the social or schedule pressure that was keeping it normalised. The story does not require that you “won” — it requires that you acted and documented.

Source: AIChE

20. Map Your Technical Background to a Specific NASA Centre, Directorate, and Active Programme — and Be Prepared to Explain What You Would Own Versus Where You Have Gaps

3–4 h

Skill: Specific self-positioning within NASA’s decentralised programme structure

Objective: NASA does not have one engineering organisation — it has ten field centres with distinct missions, cultures, and programme portfolios, plus JPL. A candidate who cannot answer “which centre, which directorate, which programme” with precision is signalling they are applying to NASA as a brand rather than to a specific engineering mission. Every vacancy announcement on USAJOBS is for a specific GS series and grade at a specific centre. The interview panel for that position consists of people working on that programme. Generic enthusiasm for space is not a substitute for knowing what you would be doing on Monday morning.

Done Signal: For the specific vacancy you are preparing for: you can name the centre, the directorate, the branch or group, and the active programme or research area most relevant to the role. You have read the centre’s most recent strategic plan or capability overview. You can name two specific technical contributions your background makes to that programme and one area where you would need to develop. You can deliver this in under 2 minutes without notes.

SUMMARY REFERENCE TABLE

#	Task Description	Target Level	Effort
1	USAJOBS-compliant resume that parses correctly	All	3–4 h
2	USA Hire practice assessments + scoring understanding	All	3–4 h
3	STAR stories mapped to vacancy-announcement competencies	All	4–5 h
4	“Why NASA?” connected to centre, programme, and problem	All	2–3 h
5	Civil servant vs. contractor distinction and implications	All	2 h
6	Domain fundamentals for target centre’s mission	All	5–6 h
7	NPR 7123.1 systems engineering framework	Mid+	5–6 h
8	Technical Authority structure and daily-work implications	Mid+	3 h
9	Formal review process story where review changed something	Mid+	2 h
10	NASA risk management (NPR 8000.4 + 5×5 matrix)	Mid+	3–4 h
11	Requirements ambiguity — documented clarification story	Mid+	2 h
12	NASA SE Handbook — trade studies, interfaces, verification	Mid+	5–6 h
13	Lessons-learned contribution story	Mid+	2–3 h
14	Independent technical authority against schedule pressure	Senior	2–3 h
15	Full programme lifecycle, all review gates	Senior	4–6 h
16	Contractor technical performance oversight story	Senior	2 h
17	PRA contribution with design or operations decision outcome	Senior	4–5 h
18	Junior engineer capability transfer — programme-specific	Senior	2 h
19	Normalisation-of-deviance story	All	2 h
20	Centre/directorate/programme self-positioning with gaps	All	3–4 h

THE STRUCTURAL REALITY

NASA is the only employer in this series where the hiring process itself is governed by federal law, where independent technical authority is a constitutional feature of the organisation rather than a cultural aspiration, and where the lessons of fourteen astronaut deaths are embedded into the governance structure every engineer works within. That is not background context — it is the job.

The single sharpest distinction between NASA and every other organisation in engineering hiring: **everywhere else, safety culture is an interview topic. At NASA, it is a governance structure with legal standing.** The Technical Authority system exists precisely because NASA learned — twice, at the cost of fourteen lives — that schedule pressure and organisational hierarchy will override individual engineers’ concerns unless the institution structurally prevents it. When NASA asks about safety at interview, it is not checking whether you have good values. It is checking whether you understand the institutional machinery that protects those values, and whether you will use it.